Breast Ultrasound Image Classification using using BUSI dataset

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AD23B1048

Introduction

Ultrasound imaging is widely used for breast cancer detection due to its safety.

This report summarizes the development and evaluation of a convolutional neural network (CNN) model for classifying breast ultrasound images into three categories: **normal**, **benign**, and **malignant**. The goal is to assist in diagnosis by automatically identifying the type of abnormality from ultrasound scans.

Dataset

- **Source**: BUSI (Breast Ultrasound Images) dataset from Kaggle.
- **Total images**: 780 grayscale images, resized to 224×224 pixels.
- **Class distribution**:
  - Normal: 133 images - No tumor
  - Benign: 437 images - Non-cancerous tumor
  - Malignant: 210 images - Cancerous tumor
- **Train/Test Split**: 80% training, 20% test, stratified by class.

    - Training set: 624 images (normal 106, benign 350, malignant 168)
    - Test set: 156 images (normal 27, benign 87, malignant 42)

The dataset is imbalanced, with benign cases being the majority.

Preprocessing

- Images were loaded in grayscale mode.
- Resized to 224×224 to match the input size expected by the CNN.
- Pixel values normalized to [0, 1].

No data augmentation was applied.

Model Architecture

A simple sequential CNN was built with the following layers:

- **Conv Block** (repeated 6 times):
    - Conv2D (8 filters, 3×3 kernel, same padding, L2 regularization 1e-4)
    - ReLU activation
    - MaxPooling2D (2×2)
- **Flatten** layer
- **Dropout** (0.5) for regularization
- **Dense** output layer with 3 units and softmax activation

Total parameters: ~ 3k. Making it small and fast.

Training

- **Loss function**: sparse categorical crossentropy

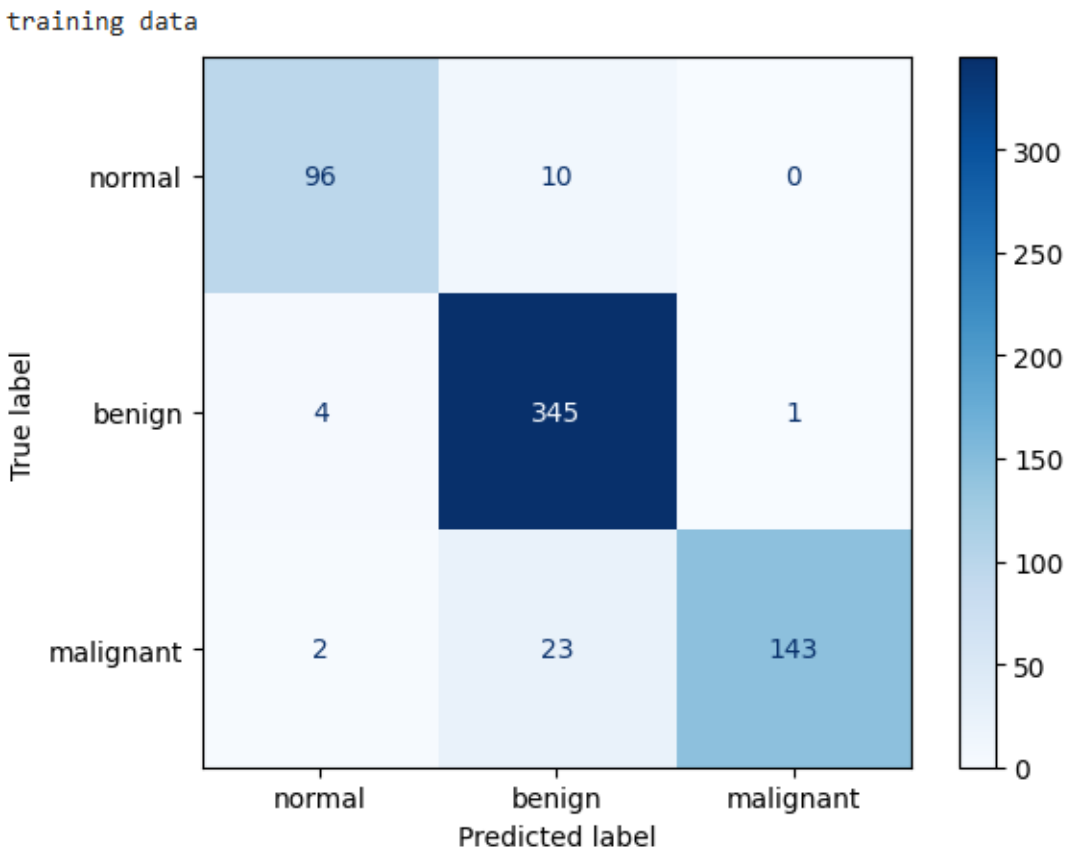
- **Optimizer**: Adam
- **Batch size**: 16
- **Epochs**: 100

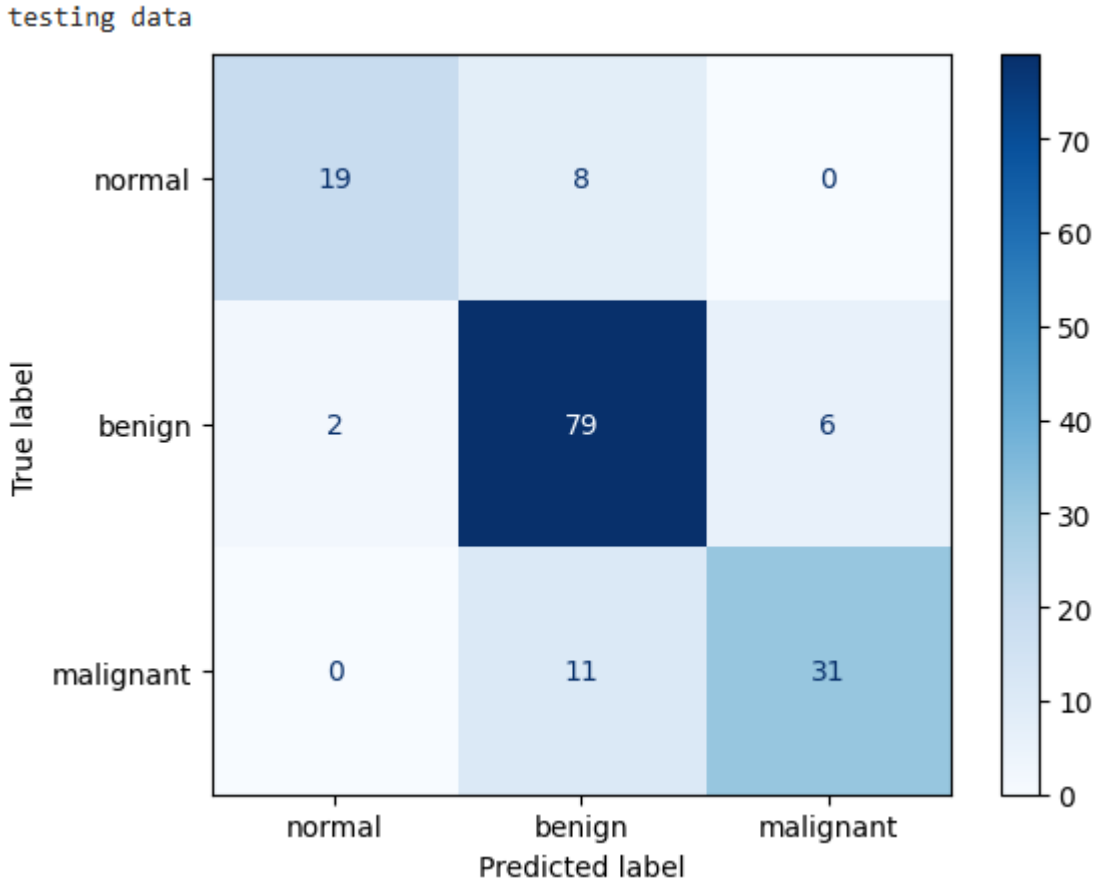
Evaluation

Training History

- The plot of training and validation accuracy shows steady improvement.
- **Best validation accuracy**: **0.8269** (achieved at epoch 95).
- Training accuracy reached ~0.82, indicating well generalization.

Per-Class Performance – confusion matrix





The model correctly classifies most normal cases. Misclassifications occur mainly between benign and malignant (6 benign predicted as malignant, 11 malignant as benign).model is not confused between normal and malignant.

Example Predictions

Ten randomly selected test images were displayed with their predicted and true labels. Nine predictions were correct, demonstrating the model's reliability on these samples.

conclusion

- The CNN achieves strong performance, with test accuracy above 82%.
- The confusion matrix reveals that the main challenge lies in distinguishing benign from malignant lesions.

- The model is lightweight (only ~3k parameters) and fast to train.
- The less discrepancy between the reported validation accuracy (0.8269) and the train accuracy (0.8241) suggests model is well regularized.